Individual-Level Evidence for Temporal Bayesian Player-Skill Estimation: A Controlled Ablation

Dhruvi Shah Divija Nayak Veera Bharakhada

CSE 516 Probabilistic Graphical Models | Group 3 | Milestone 4 (Final Submission)

Abstract. In team sports a single match outcome is shared by teammates, so it carries weak evidence about each player's individual ability. We study whether adding individual-level performance observations to a temporal Bayesian network of latent player skills improves recovery of individual skill. The model is a simplified TrueSkill-Through-Time-style graph: Gaussian random-walk skills, a logistic team-outcome likelihood, and (in the extension) Gaussian individual observations, fitted by MAP with a Laplace uncertainty approximation. Under a controlled synthetic doubles generator with known ground truth, and ten paired seeds (42–51) evaluated chronologically (train $t \le 20$, held out $t \ge 21$), the extension reduces skill MAE from 0.4372 ± 0.1400 to 0.2632 ± 0.0366 (mean paired reduction 0.1740; 95% t-CI $[0.066, 0.282]$; bootstrap CI $[0.097, 0.274]$; paired t-test $p = 0.0055$; improvement in 10/10 seeds). Held-out log-loss (NLL) improves from 0.5835 to 0.5601 and Brier score from 0.1999 to 0.1893; all four KPI improvements remain significant after Holm correction $p < 0.025$. The gain is large for skill recovery, small for outcome prediction, and shrinks as individual observations become noisier or sparser. All data are synthetic: this is a proof-of-concept, not a real-ATP validation, not a reproduction of full TrueSkill Through Time, and not a benchmark against the published state of the art.

1 Introduction

Skill-rating systems such as Elo, TrueSkill and TrueSkill Through Time (TTT) infer latent abilities from match results [2, 3]. In doubles or team play, however, the observed result Y depends only on the *combined* strength of each side. Two teammates who always play together are statistically hard to separate, and a win credits (or blames) both equally. The practical question we ask is narrow and testable: *if some individual-level evidence is available for each participant, how much does it help to disentangle teammates?*

Project evolution. Milestone 1 (M1) proposed a broad study of inference choices and temporal smoothing in TTT. M2 built a temporal-PGM technical foundation and a separate M2 extension identified the shared-outcome identifiability weakness in doubles as the limitation worth testing. M3 implemented and evaluated the resulting ablation, and this report (M4) consolidates it. M1 directions that were not carried through are *not* claimed as contributions (Section 5).

Contributions. (i) A clear graphical-model formulation of a team-outcome-only baseline and an individual-evidence extension that share the same temporal structure; (ii) a controlled, paired, chronological ablation over ten seeds with KPIs for skill recovery and for probability quality; (iii) robustness sweeps over match count, missing observations and observation noise, with a player-level failure analysis; (iv) an honest positioning against the TTT anchor and an explicit list of what the study does not show.

2 Problem Formulation

Let $i \in \{1, \dots, P\}$ index players ($P = 12$) and $t \in \{0, \dots, T-1\}$ index time points ($T = 30$). Each player has a latent skill $S_{i,t} \in \mathbb{R}$. At each t, $M = 12$ doubles matches m are played; match m has teams A_m, B_m of two distinct players each and a binary outcome $Y_m = 1$ if A_m wins. In the extension each participating player also produces an individual observation $Z_{i,m}$ (a synthetic performance score).

Inference target. The time-indexed skill posterior $p(S_{1:P,0:T-1} \mid \mathscr{D}_{\text{train}})$ given evidence from $t \le 20$ only. **Prediction target.** The win probability $P(Y_m = 1)$ for held-out matches at $t \ge 21$. **Hypothesis.** Adding Z to Y under otherwise identical conditions lowers the error of the recovered skills against the known synthetic truth. **Scope.** Structure is fixed from player participation and time ordering; no structure learning is performed and hyperparameters are set by the experiment, not learned.

3 Background

Why a PGM. Skills are latent, evolve over time, interact through shared matches and are observed through noisy, heterogeneous evidence. A directed graphical model expresses each of these as an explicit factor, makes the conditional-independence assumptions auditable, and gives a principled posterior with uncertainty [4]. Adding a new evidence source is a local change to the graph (one extra factor family), which is exactly what a clean ablation requires.

Temporal skill model. Each skill follows a first-order Gaussian random walk, so information about $S_{i,t}$ propagates to neighbouring times through the chain; this is the "through time" idea of TTT. **Team likelihood.** We use a logistic link on the difference of team-average skills (Section 6). **Approximate inference.** With a non-conjugate logistic likelihood the exact posterior is intractable; we use MAP estimation with a Laplace (local Gaussian) approximation around the mode [5].

4 Related Work

TrueSkill models player skills as Gaussians and infers them by message passing on a factor graph [2]. TrueSkill Through Time adds temporal dynamics so that skill estimates become comparable across eras, again through message passing [3]. The recent JSS paper of Landfried and Mocskos [1] provides a reliable, documented TTT package (Julia, Python, R) with attention to initial skill estimates and historical comparability. Our work is a smaller, different question inside this family: not a new inference algorithm, but an ablation of *what evidence enters the graph*. Probability quality is assessed with the proper scoring rules log-loss and Brier score [7].

5 SOTA Positioning

Anchor: Landfried and Mocskos (2025) [1], *TrueSkill Through Time: Reliable Initial Skill Estimates and Historical Comparability with Julia, Python, and R*, J. Stat. Softw. 112(6), doi:10.18637/jss.v112.i06. Following the course loop (Identify, Understand, Compare, Validate, Reposition):

Step	Where this project stands
Identify	Anchor is the published TTT system [1], the published reference implementation of TTT (Julia, Python, R).
Understand	TTT keeps a temporal graphical model of player skills and propagates evidence by message passing / analytical approximation. We borrow the temporal latent-skill idea only.
Compare	Ours: simplified logistic team likelihood, MAP fitting, Laplace uncertainty, fixed hyperparameters, 12 synthetic players. Anchor: full published formulation and EP-style inference, designed for real histories.
Validate	*No head-to-head run against the anchor was performed.* The only validated claim is the internal ablation (baseline vs. extension) on synthetic data.
Reposition	M1 aimed at a broader study of TTT inference and smoothing. The final claim is narrower: a proof-of-concept that individual evidence helps identify teammates in a simplified temporal PGM. We make **no** claim of superiority over, or equivalence to, the published TTT implementation.

6 Graphical Representation

Variables. $S_{i,t}$ latent skill; $Y_m \in \{0,1\}$ team outcome (observed); $Z_{i,m}$ individual observation (observed, extension only). **Generative model** (hyperparameters: $\sigma_0 = 1$, $\sigma_d = 0.12$, $\tau = 0.8$, $\sigma_z = 0.65$):

$$S_{i,0} \sim \mathcal{N}(0, \sigma_0^2), \quad S_{i,t} \mid S_{i,t-1} \sim \mathcal{N}(S_{i,t-1}, \sigma_d^2), \quad (1)$$

$$Y_m \mid S \sim \text{Bernoulli}\big(\sigma(\eta_m)\big),\ \eta_m = \tfrac{\bar{S}_{A_m} - \bar{S}_{B_m}}{\tau}, \quad (2)$$

$$Z_{i,m} \mid S_{i,t} \sim \mathcal{N}(S_{i,t}, \sigma_z^2). \quad (3)$$

Here $\bar{S}_A$ is the mean skill of the two members of team A at the match time and $\sigma(\cdot)$ the logistic function.

Factorisation. With $\mathscr{M}_{\text{tr}}$ the training matches and $\mathscr{O}$ the available individual observations,

$$p(S,Y,Z) = \prod_i \Big[p(S_{i,0}) \prod_{t>0} p(S_{i,t} \mid S_{i,t-1}) \Big] \cdot \prod_{m \in \mathscr{M}_{\text{tr}}} p(Y_m \mid S_{\text{teams}(m)}) \cdot \prod_{k \in \mathscr{O}} p(Z_k \mid S_{\text{pl}(k),\text{t}(k)}), \quad (4)$$

and the baseline drops the last product. **Conditional independencies** (modelling assumptions, not empirical causal facts): (a) skills are Markov in time; (b) given the skills of the four participants, Y_m is independent of all other matches; (c) given its own skill, Z is independent of Y and of other observations. The shared-outcome node Y_m has the two teammates as co-parents: observing Y_m couples them (explaining-away structure), which is the source of the identifiability problem, and Z provides separate, player-specific paths into each skill.

7 Learning Method

The graph and all hyperparameters are fixed by the experiment, so **parameter learning** here means estimating the latent trajectories for fixed $(\sigma_0, \sigma_d, \tau, \sigma_z)$, not learning those hyperparameters, and no **structure learning** is performed (hence structure metrics such as SHD do not apply). The skills are the posterior mode,

$$\hat{S} = \arg\min_S \tfrac{1}{2} S^\top Q S + \sum_{m \in \mathscr{M}_{\text{tr}}} \big[\log(1 + e^{\eta_m}) - y_m \eta_m \big] + \sum_{k \in \mathscr{O}} \frac{(S_{\text{pl}(k),\text{t}(k)} - z_k)^2}{2\sigma_z^2}, \quad (5)$$

where Q is the precision matrix of the Gaussian random-walk prior and the last term is dropped for the baseline. This is a smooth convex problem (a quadratic prior plus logistic and quadratic terms), solved with SciPy L-BFGS-B [6] (`maxiter`=500, `ftol`=10^{-9}) from $S = 0$ using analytic gradients; success flags and iteration counts are logged for every run (all runs converged successfully in the archived CSV). We justify MAP over sampling or EP for this study because it is fast, deterministic given the data, and identical across the two conditions, so the *only* difference between arms is the evidence.

8 Inference Method

Inference problem. Estimate the posterior mode (MAP) and approximate posterior uncertainty of Si,t for all (i,t), then predict held-out outcomes.

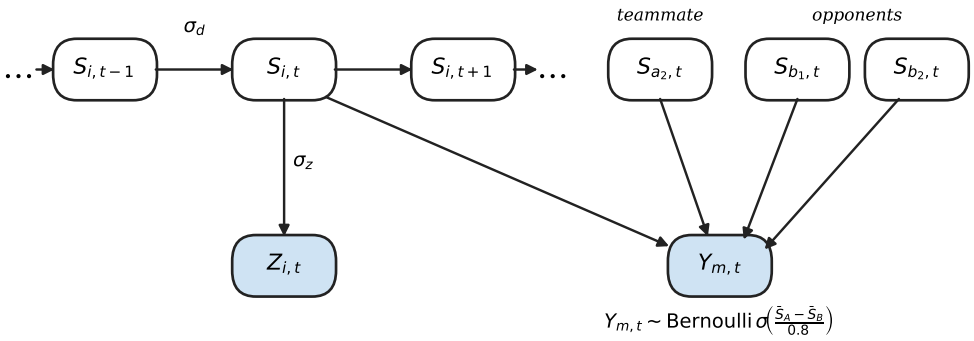
tion: the covariance is H^{-1} with $H = Q + \sum_m w_m s_m s_m^\top + \sum_k \sigma_z^{-2} e_k e_k^\top$, where $w_m = \sigma(\eta_m)(1 - \sigma(\eta_m))/(4\tau^2)$ and $s_m = (+1, +1, -1, -1)$ on the four participants of match m (derivation in Appendix A); a 10^{-8} diagonal jitter is added and the posterior SD is $\sqrt{\text{diag}\, H^{-1}}$. (3) Held-out win probability $\sigma\big((\hat{\bar{S}}_A - \hat{\bar{S}}_B)/\tau\big)$ at the match time, with *no* conditioning on test outcomes or test observations.

Held-out semantics. Because no evidence exists at $t \geq 21$, the MAP trajectory over those times is the prior chain extended from $t = 20$ (a flat continuation). Test-period skill MAE/RMSE therefore measure *forward extrapolation* against simulated truth, and the predictive KPIs measure genuine chronological held-out prediction; neither is retrospective smoothing with test evidence.

Computational trade-off. The state has $d = PT = 360$ dimensions. Each L-BFGS-B iteration costs one pass over matches and observations; the Laplace step builds a dense $d \times d$ Hessian and inverts it ($O(d^3)$, trivial at $d = 360$ but a bottleneck for large rosters, where sparse/banded solves or EP would be needed). The posterior SD is only a local Gaussian approximation: smaller SD is not evidence of calibrated credible intervals.

9 Experimental Setup

Data. The ATP match history referenced in earlier milestones lacks the individual-level statistics needed for this question, so the evaluation uses **synthetic doubles data**: for each seed the generator draws initial skills, random-walk increments, four



Shaded = observed (Z only in the extension; Y in both models). S: latent skill, Z: individual observation, Y: shared team outcome. Player i is shown as the focal member a_1 of team A; the plate repeats over players i, times t and matches m at each t.

Figure 1: Graphical model (schematic). Skill chains $S_{i,0:T-1}$ are linked through shared outcome nodes $Y_{m,t}$; individual observations $Z_{i,t}$ (extension only) are conditionally independent given the player's skill. The baseline omits Z.

distinct random players per match, a Bernoulli outcome from the logistic team-average difference, and $Z = S + \mathcal{N}(0, \sigma_z^2)$ for each participant. No real-world individual observations are used or claimed. **Protocol.** 12 players, 30 time points, 12 matches/time (360 matches/seed), chronological 70:30 split (train $t = 0$–20: 252 matches; held out $t = 21$–29: 108 matches), seeds 42–51. Both arms see the *same* generated data per seed (paired design); neither test outcomes nor test observations enter fitting. Robustness sweeps vary matches per time $\{6, 12, 24\}$, missing-observation rate $\{0, 0.25, 0.5, 0.75\}$, and observation noise $\sigma_z \in \{0.25, 0.65, 1.0, 1.5\}$ (the model's σ_z is matched to the generator's). **Reproducibility.** Code, seeds and commands are in Appendix B.

10 Baselines

Baseline (team-only). The identical temporal graph, prior, optimiser and split, but with the individual-observation factors removed: the model sees only Y_m. **Proposed (extension).** The same plus the Z factors. Because the two arms differ in exactly one evidence source, any difference is attributable to that source. The published TTT system was *not* implemented or run as a baseline (Section 5).

11 KPIs

Skill recovery (primary): MAE and RMSE of $\hat{S}_{i,t}$ against the known truth over held-out times $t \geq 21$. **Probability quality (secondary), on held-out outcomes:** log-loss $= -\frac{1}{n}\sum[y \log p + (1-y)\log(1-p)]$ (equal to the Bernoulli negative log-likelihood per observation, NLL), Brier score $= \frac{1}{\cdot}\sum(p-y)^2$, and a reliability diagram for calibration. **Com-**

Lower prediction errors and computational costs are preferable; smaller posterior SD does not necessarily imply better calibration. Statistics: mean ± sample SD over seeds, 95% t-intervals (t0.975,9 = 2.262), paired t-test, Wilcoxon signed-rank test, percentile bootstrap

primary/secondary KPIs, and paired effect size as the mean paired difference divided by the SD of paired differences.

12 Results

Table 1 and Figure 2 place the baseline and extension side by side under identical conditions. The extension lowers skill MAE and RMSE by about 40% and 39% relative to the baseline, lowers held-out NLL and Brier by smaller amounts, converges in 2.4× fewer iterations, and has much smaller posterior SD. The baseline's across-seed SD of MAE (0.140) is about four times the extension's (0.037): without individual evidence recovery is not only worse but far less stable (e.g. seed 47: 0.745 vs. 0.226).

Robustness

The MAE advantage persists in every swept configuration (Table 2; every paired $p \leq 0.016$; the 12-match, 0%-missing, $\sigma_z = 0.65$ cell is the main experiment). It narrows monotonically as missingness rises ($\bar{\Delta} = 0.174 \to 0.144$ at 75% missing) and as observation noise rises ($0.201 \to 0.134$ at $\sigma_z = 1.5$), and it does not vanish at 75% missing observations because teammates still receive some individual evidence. At 6 matches per time the NLL gain is not significant ($p = 0.25$) even though the skill gain is ($p < 0.001$).

Error analysis

Improvement is not uniform: in seed 42 the extension reduces held-out MAE for 8 of 12 players (large gains for players 2, 5, 7, 9, 11) but is *worse* for four (players 0, 1, 4, 10; e.g. player 0: $0.217 \to 0.465$). This is a single-seed descriptive analysis, so we draw no population-level inference from which players degrade.

13 Statistical Analysis

All four KPI differences favour the extension and are significant after Holm correction (Table 3). For the primary KPI, the mean paired MAE reduction is 0.1740 (paired $t_9 = 3.632$, $p = 0.0055$; Wilcoxon $p = 0.0020$, the smallest attainable with $n = 10$; $d_z = 1.15$; 10/10 seeds improve). The percentile-bootstrap interval $[0.097, 0.274]$ reported in M3 is narrower than the t-interval $[0.066, 0.282]$; with $n = 10$ and one extreme

Table 1: Final KPIs, ten paired seeds (42–51): mean ± sample SD [95% t-interval of the mean]. Skill KPIs are vs. simulated truth on held-out times; NLL/Brier on held-out matches; posterior SD averaged over held-out times. Runtime is environment-dependent.

KPI	Team-only baseline	+ Individual evidence
Skill MAE ↓	0.4372 ± 0.1400 [0.3370, 0.5374]	0.2632 ± 0.0366 [0.2371, 0.2894]
Skill RMSE ↓	0.5357 ± 0.1586 [0.4222, 0.6492]	0.3284 ± 0.0395 [0.3002, 0.3567]
Log-loss / NLL ↓	0.5835 ± 0.0568 [0.5428, 0.6241]	0.5601 ± 0.0655 [0.5132, 0.6070]
Brier score ↓	0.1999 ± 0.0242 [0.1826, 0.2172]	0.1893 ± 0.0269 [0.1700, 0.2086]
Runtime (s) ↓	2.090 ± 0.865 [1.471, 2.708]	1.470 ± 0.376 [1.201, 1.739]
L-BFGS-B iterations ↓	164.5 ± 12.6 [155.5, 173.5]	67.4 ± 4.2 [64.4, 70.4]
Mean posterior SD ↓	0.6182 ± 0.0066 [0.6135, 0.6229]	0.3182 ± 0.0005 [0.3178, 0.3186]

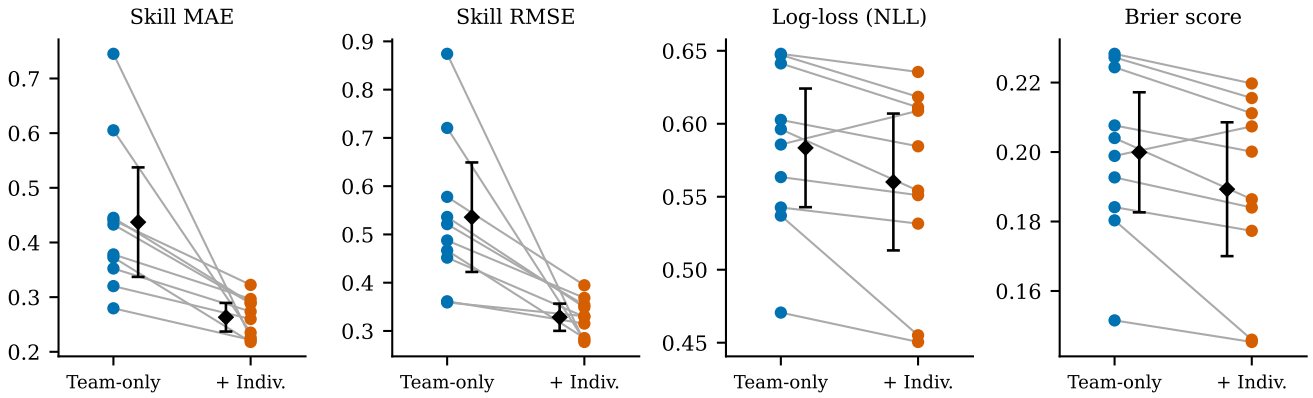


Figure 2: Paired per-seed KPIs. Grey lines join the two arms for the same seed; black diamonds show the mean with 95% t-interval. The extension improves skill MAE and RMSE in 10/10 seeds and NLL and Brier in 9/10.

The bootstrap interval is narrower than the t-interval; both exclude zero, but coverage cannot be established from these ten seeds.

Effects on probability quality are statistically supported but small in absolute terms (NLL −0.0234, Brier −0.0106), with d_z of 0.87 and 0.99.

Probability quality and calibration. Log-loss (NLL) and Brier score are proper scoring rules; both improve. The reliability diagram (Figure 4, right) is a descriptive single-seed diagnostic with few matches per bin (108 held-out matches over 8 bins); both models track the diagonal roughly, with deviations in the low-probability bins, and the diagram does not establish global calibration. A scalar calibration error (ECE) was not computed in the archived results. **Caveats.** Seeds are independent simulations of a single data-generating process, so intervals describe simulation variability, not uncertainty about real data; with only ten seeds, effect-size estimates are themselves imprecise; the paired design is essential because the baseline's across-seed variance is large.

14 Discussion

What was learned. The main finding is an *identification benefit*: individual-specific observations separate teammates whose shared result cannot. Skill recovery improves strongly, while outcome prediction improves only modestly. A plausible, though not separately tested, explanation: the held-out trajectory is a flat extrapolation from $t = 20$, and team win probability depends on a *difference of team averages*, in which the ambiguity between teammates partly cancels, so prediction benefits less than individual estimation. **Why iterations and posterior SD fall.** The extra factors add per-skill curvature, which plausibly makes the objective better conditioned (fewer iterations) and certainly tightens the Laplace covariance; we caution that smaller approximate SD is not demonstrated calibration. **Failures.** Gains shrink with noise and missingness (Figure 3) and are not uniform across players (Figure 4). **Robustness to the baseline's spread.** The baseline's high seed variance (some seeds with MAE > 0.6) indicates that team-only recovery is sensitive to which teammates happen to be paired often.

15 Limitations

(1) The individual evidence is generated directly from ground-truth skill, so it is informative by construction; real observations may be biased, context-dependent, correlated with outcomes or measure different latent traits. (2) All data are synthetic with 12 players; there is no real individual-statistics validation. (3) The model is simplified: logistic team likelihood, Gaussian random-walk, fixed hyperparameters, MAP plus Laplace rather than full TTT with message passing; the comparison to the anchor is conceptual, not empirical. (4) The model's σ_z equals the generator's, a favourable well-specified setting; mis-specification was not tested. (5) Ten seeds, one generator; no calibration scalar; one-seed error analysis. (6) Runtimes depend on hardware and differ across

Table 2: Robustness sweeps (ten seeds each; MAE mean ± SD; $\bar{\Delta}$ = team-only − extension with 95% t-CI; NLL difference). The team-only arm does not use Z, so it is unchanged across the missing-rate and noise sweeps. Different match counts generate different datasets, so team-only MAE need not be monotone in match count.

Sweep	Level	Team-only MAE	+ Indiv. MAE	$\bar{\Delta}$MAE [95% CI]	p_t	$\bar{\Delta}$NLL	p_t
Matches / time	6	0.5037 ± 0.0943	0.2989 ± 0.0378	0.2047 [0.126, 0.284]	<0.001	0.0204	0.252
	12	0.4372 ± 0.1400	0.2632 ± 0.0366	0.1740 [0.066, 0.282]	0.005	0.0234	0.022
	24	0.4527 ± 0.1354	0.2392 ± 0.0378	0.2135 [0.118, 0.309]	<0.001	0.0207	0.007
Missing rate	0	0.4372 ± 0.1400	0.2632 ± 0.0366	0.1740 [0.066, 0.282]	0.005	0.0234	0.022
	0.25	0.4372 ± 0.1400	0.2656 ± 0.0329	0.1716 [0.061, 0.282]	0.007	0.0231	0.018
	0.5	0.4372 ± 0.1400	0.2740 ± 0.0268	0.1632 [0.066, 0.260]	0.004	0.0226	0.020
	0.75	0.4372 ± 0.1400	0.2937 ± 0.0280	0.1435 [0.045, 0.242]	0.009	0.0223	0.013
Obs. noise σ_z	0.25	0.4372 ± 0.1400	0.2358 ± 0.0408	0.2014 [0.096, 0.307]	0.002	0.0254	0.019
	0.65	0.4372 ± 0.1400	0.2632 ± 0.0366	0.1740 [0.066, 0.282]	0.005	0.0234	0.022
	1	0.4372 ± 0.1400	0.2822 ± 0.0351	0.1550 [0.049, 0.261]	0.009	0.0216	0.024
	1.5	0.4372 ± 0.1400	0.3028 ± 0.0403	0.1344 [0.032, 0.237]	0.016	0.0193	0.025

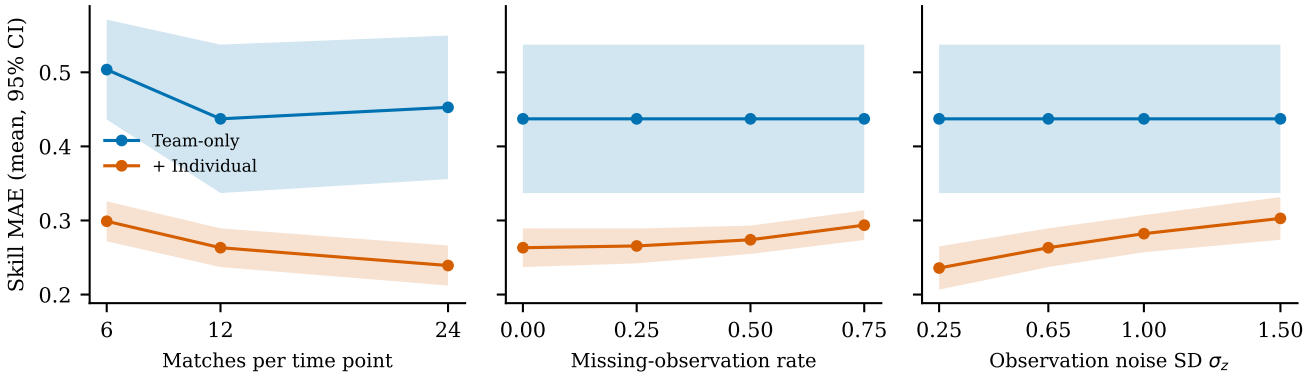


Figure 3: Skill MAE under sample-size, missing-observation and observation-noise sweeps (mean and 95% t-interval). The extension advantage persists across all settings but narrows with sparser or noisier individual evidence.

archived runs (e.g. the same 12-match configuration is timed separately in the main and sweep experiments). (7) Dense $O(d^3)$ covariance does not scale to large rosters.

Next research step. Obtain real individual-level doubles performance features, derive an empirically justified observation likelihood (including bias and missingness mechanisms), learn rather than fix σ_d, τ, σ_z, run the same chronological evaluation on real matches with calibration error and subgroup/player-level failure checks, and finally benchmark against the published TTT implementation on identical data.

16 Conclusion

In a controlled, paired, chronological synthetic study, adding individual-level observations to a temporal Bayesian skill model reduced held-out skill MAE by 40% (10/10 seeds, Holm-corrected $p = 0.020$) and improved NLL and Brier modestly (Holm $p = 0.024$), with the benefit persisting but shrinking under noise and missingness. The result supports a narrow ablation hypothesis in a favourable synthetic setting. Relative to the TTT anchor the project remains a proof-of-concept; whether the benefit transfers to real tennis data is open.

References

[1] G. Landfried and E. Mocskos. TrueSkill Through Time: Reliable Initial Skill Estimates and Historical Comparability with Julia, Python, and R. *Journal of Statistical Software*, 112(6), 2025. doi:10.18637/jss.v112.i06.

[2] R. Herbrich, T. Minka, T. Graepel. TrueSkill: A Bayesian skill rating system. *NIPS 19*, 2006.

[3] P. Dangauthier, R. Herbrich, T. Minka, T. Graepel. TrueSkill Through Time: Revisiting the history of chess. *NIPS 20*, 2007.

[4] D. Koller and N. Friedman. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press, 2009.

[5] C. M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006 (Laplace approximation, Sec. 4.4).

[6] R. H. Byrd, P. Lu, J. Nocedal, C. Zhu. A limited memory algorithm for bound constrained optimization. *SIAM J. Sci. Comput.*, 16(5):1190–1208, 1995.

[7] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

A Derivations

Write $\eta_m = s_m^\top S_{(m)}/(2\tau)$ with $s_m = (+1,+1,-1,-1)$ over the four participants of match m. The negative log joint contribu-

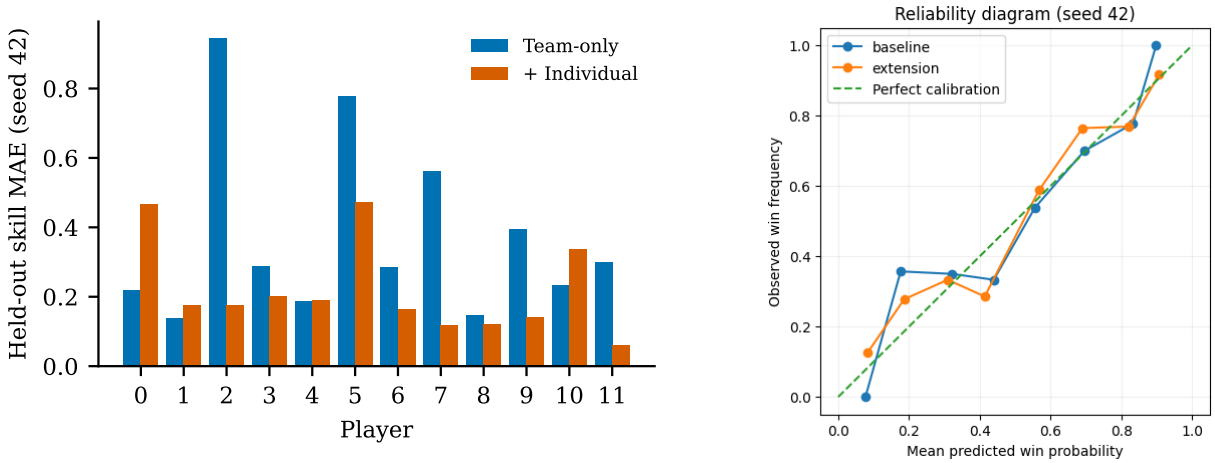


Figure 4: Left: per-player held-out skill MAE, seed 42 (extension is worse for players 0, 1, 4, 10). Right: reliability diagram (seed 42, 8 bins, 108 held-out matches) for both models.

Table 3: Paired differences Δ = baseline − extension (positive favours the extension), $n = 10$ seeds. Holm correction over the four KPI tests.

KPI	$\bar{\Delta}$	95% t-CI	95% boot. CI	t_9	p_t	p_{Wilcoxon}	p_{Holm}
Skill MAE	0.1740	[0.0656, 0.2823]	[0.0971, 0.2736]	3.632	0.005	0.002	0.020
Skill RMSE	0.2073	[0.0806, 0.3339]	[0.1140, 0.3198]	3.702	0.005	0.002	0.020
Log-loss	0.0234	[0.0042, 0.0425]	[0.0085, 0.0400]	2.762	0.022	0.027	0.024
Brier	0.0106	[0.0029, 0.0183]	[0.0048, 0.0173]	3.128	0.012	0.014	0.024

tion of the outcome is $\ell_m = \log(1 + e^{\eta_m}) - y_m \eta_m$, so

$$\frac{\partial \ell_m}{\partial S_j} = \frac{\sigma(\eta_m) - y_m}{2\tau} s_{m,j},$$
$$\frac{\partial^2 \ell_m}{\partial S_j \partial S_l} = \frac{\sigma(\eta_m)\big(1 - \sigma(\eta_m)\big)}{4\tau^2} s_{m,j} s_{m,l}.$$

The prior contributes QS to the gradient and Q to the Hessian, where $S_{i,0}$ has precision σ_0^{-2} and each increment adds the tridiagonal block $\sigma_d^{-2} \left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right]$. Each observed Z_k contributes $(S_{p,t} - z_k)/\sigma_z^2$ to the gradient and σ_z^{-2} to the Hessian diagonal. At the mode the Laplace covariance is $\Sigma \approx (H + \varepsilon I)^{-1}$ with $\varepsilon = 10^{-8}$. Because ℓ_m is convex in S and the other terms are quadratic with $Q \succeq 0$ plus an anchor at $t = 0$, the objective is strictly convex and the mode is unique. When no evidence exists at a time $t \geq 21$, the objective in those variables is purely the quadratic increment penalty, minimised by a constant continuation of $S_{\cdot,20}$.

B Reproducibility

Package layout (`M4_G3/`): `Report/`, `Presentation/`, `Video/`, `Code/src/` (canonical notebook `CSE516_M3_Results_Analysis.ipynb`: generator, baseline, extension, MAP/Laplace inference, all experiments),

`Code/verify_results.py`, `Code/requirements.txt` (numpy, pandas, scipy, matplotlib, jupyter), `Data/` (dataset description and an example `synthetic_doubles_data.csv`), `Results/` (per-seed CSV, summaries, robustness CSVs, paired-test JSON, plots), and `Supplementary/` (M1, M2, M3 archives, not additional M4 experiments).

Commands (from `M4_G3/`):
`python -m pip install -r Code/requirements.txt`
`python Code/verify_results.py` (checks the archived 20 rows, seeds 42–51, and prints mean ± SD and the paired test)
`cd Results && jupyter notebook ../Code/src/CSE516_M3_Results_Analysis.ipynb` (run top to bottom to regenerate; output paths are relative to `Results/`).
Settings. Seeds 42–51 (generator uses `numpy.random.default_rng(seed)`; bootstrap seed 2026); $P = 12$, $T = 30$, $M = 12$, $\sigma_0 = 1$, $\sigma_d = 0.12$, $\tau = 0.8$, $\sigma_z = 0.65$; split at $t = 21$; L-BFGS-B `maxiter`=500, `ftol`=10^{-9}. Runtimes vary by machine and SciPy version. **Provenance.** The archived per-seed CSV is the numerical source of truth. All numbers in this report are recomputed from the archived CSV files; the t-intervals, Wilcoxon tests, Holm corrections, d_z values and sweep statistics in Tables 1–2 were derived from those archived per-seed results, and no new simulation was run for M4. The M3 notebook itself does not print those additional statistics, only the paired t, p and the bootstrap CI.

C Per-seed results

Table 4: Archived per-seed held-out KPIs (Team = team-only baseline, +Ind. = extension), lower is better.

	MAE		RMSE		Log-loss		Brier	
Seed	Team	+Ind.	Team	+Ind.	Team	+Ind.	Team	+Ind.
42	0.3727	0.2180	0.4668	0.2853	0.5962	0.5542	0.2040	0.1864
43	0.2796	0.2218	0.3615	0.3153	0.4706	0.4505	0.1515	0.1453
44	0.6052	0.2351	0.7207	0.2806	0.6472	0.6184	0.2272	0.2155
45	0.3524	0.2737	0.4519	0.3294	0.6026	0.5846	0.2077	0.2001
46	0.3782	0.2966	0.4875	0.3685	0.5426	0.5316	0.1841	0.1773
47	0.7451	0.2260	0.8742	0.2772	0.5371	0.4551	0.1803	0.1460
48	0.4328	0.2886	0.5213	0.3538	0.6479	0.6355	0.2283	0.2197
49	0.4451	0.2901	0.5362	0.3490	0.5635	0.5511	0.1927	0.1840
50	0.3202	0.2600	0.3591	0.3307	0.5858	0.6089	0.1989	0.2074
51	0.4407	0.3224	0.5778	0.3947	0.6413	0.6113	0.2244	0.2112